ACTIVE DIRECTORY

PROFESSIONAL REPORT

Design, Implementation, Management and Security Assessment of **Active Directory** Services



CENTRALIZED MANAGEMENT
Efficient management of users, computers, and resources across the organization.

SECURITY & COMPLIANCE
Enforce policies, secure access, and ensure compliance with industry standards.

RELIABILITY & SCALABILITY
Deliver high availability and scalability to support business continuity and growth.

Assessment scope 2
Summary 3
Overview: 3
Summary: 3
Impact: 3
Remediation direction: 3
Vulnerability writeups 4
Title: 4
Risk Rating: 4
Summary: 4
Background: 4
Technical Details and Evidence: 4
Impact: 5
Remediation advises: 6
Title: 6
Risk rating 6
Summary 6
Background: 6
Impact: 7
Remediation advises: 7
Title: 7
Risk rating: 7
Summary: 7
Background: 7
Technical Details and Evidence: 7
Impact: 9
Remediation advises: 9
Title: 10
Risk rating 10
Summary: 10
Background: 10

Technical Details and Evidence: 10
Impact: 11
Remediation advises: 11
Risk rating 11
Summary: 11
Background: 11
Technical Details and Evidence: 12
Impact: 14
Remediation advises: 14
Title 14
Risk rating 14
Summary: 14
Background: 15
Technical Details and Evidence: 15
Impact: 16
Remediation advises: 16

Assessment scope

The assessment covered the thm.local Active Directory environment and focused on identifying security weaknesses that could lead to unauthorized access or privilege escalation. The assessment included network and service enumeration, SMB and LDAP configuration, Kerberos authentication, user and permission enumeration, credential security, RDP access, and privilege escalation. Testing aimed to identify vulnerabilities, validate their impact, and determine whether an attacker could progress from initial access to privileged system access, with successful access to flag.txt.txt used as proof of impact.

Summary

Overview:

A security assessment was conducted against the thm.local Active Directory environment to identify vulnerabilities that could enable unauthorized access and privilege escalation. The assessment covered domain enumeration, SMB/LDAP services, Kerberos authentication, user permissions, credential security, and privilege escalation paths.

Multiple weaknesses were identified and successfully chained together, ultimately resulting in system level access to the Domain Controller.

Summary:

The assessment identified multiple security weaknesses, including weak credentials, password reuse, excessive user permissions, and exposed credentials in a PowerShell script. These vulnerabilities were successfully chained to obtain system level access to the Domain Controller.

Impact:

The identified weaknesses could allow an attacker to compromise multiple domain accounts, escalate privileges, and leds to system level access to the Domain Controller. This results in unauthorized access to sensitive data, account compromise, and potential full control of the Active Directory environment.

Remediation direction:

The environment should be secured by removing unnecessary Guest access, enforcing strong unique passwords, and reviewing excessive Active Directory permissions.

Plaintext credentials should be removed from scripts, and privileged accounts should follow least-privilege principles. Regular monitoring and AD security reviews should also be implemented to prevent similar attacks

Vulnerability writeups

Title: Weak authentication that allows guest accounts enabled with no password.

Risk Rating: High (CVSS score-7,8)

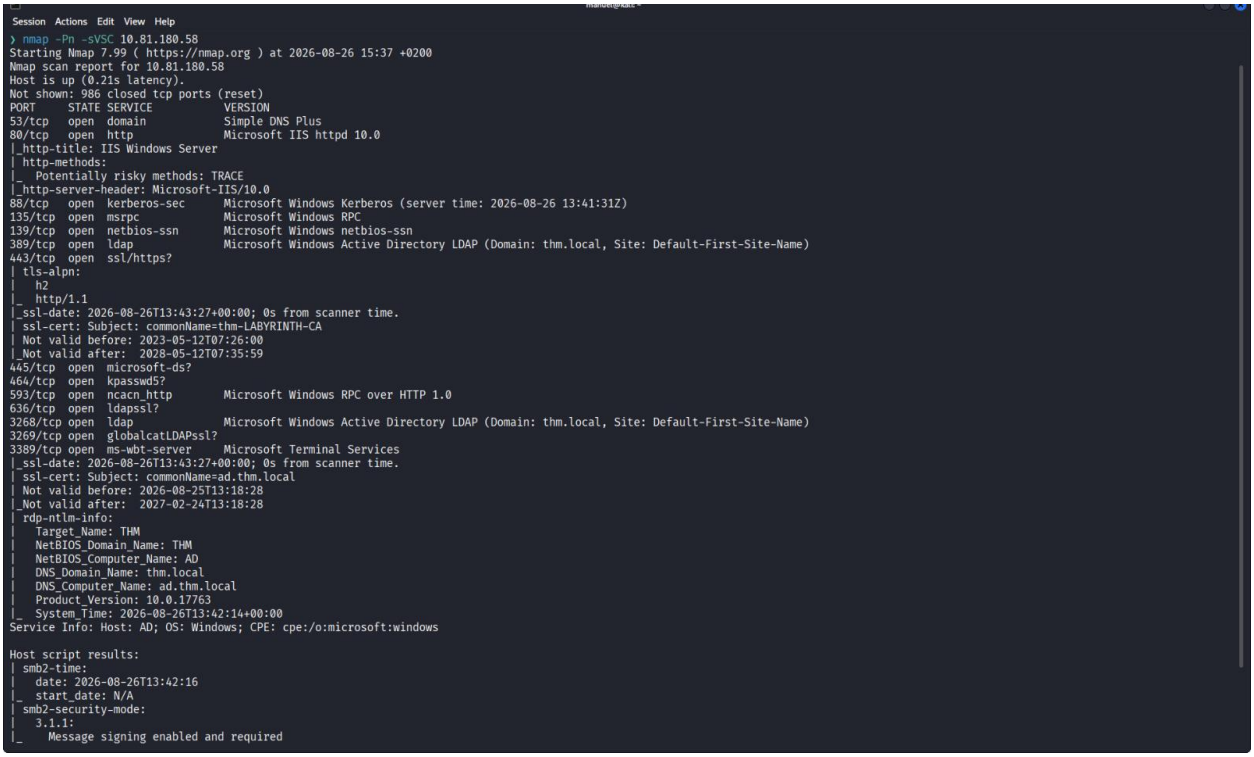
Summary:

The Guest account was enabled and allowed authentication using a blank password. This provided an initial foothold that could be used to enumerate Active Directory resources and identify further attack paths.

Background:

An attacker who discovers an enabled Guest account with a blank password could use it as an initial foothold into the domain environment. Depending on the account's permissions, they may be able to enumerate users, groups, shares, and other Active Directory information, helping them identify valid accounts and potential paths for further compromise.

Technical Details and Evidence:

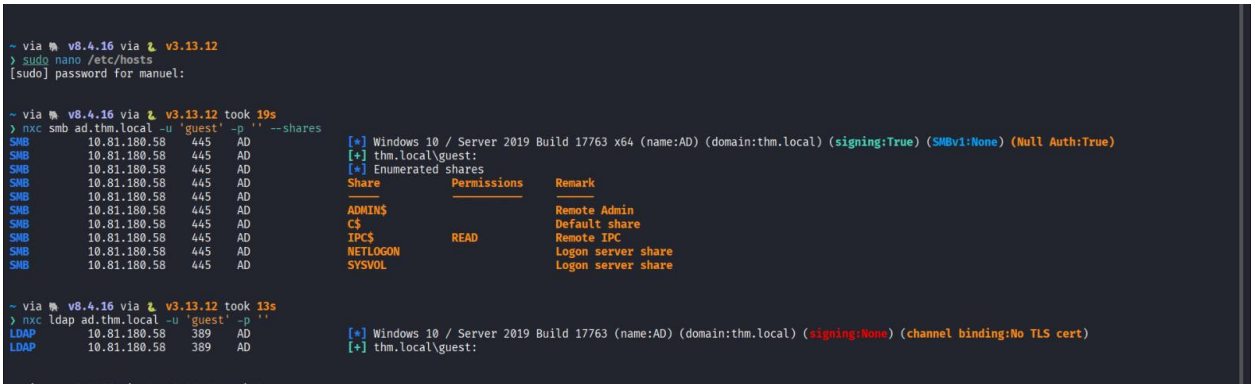


```
Session  Actions  Edit  View  Help
❯ nmap -Pn -sVSC 10.81.180.58
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 15:37 +0200
Nmap scan report for 10.81.180.58
Host is up (0.21s latency).
Not shown: 986 closed tcp ports (reset)
PORT     STATE SERVICE          VERSION
53/tcp   open  domain           Simple DNS Plus
80/tcp   open  http             Microsoft IIS httpd 10.0
|_http-title: IIS Windows Server
| http-methods:
|_  Potentially risky methods: TRACE
|_http-server-header: Microsoft-IIS/10.0
88/tcp   open  kerberos-sec     Microsoft Windows Kerberos (server time: 2026-08-26 13:41:31Z)
135/tcp  open  msrpc            Microsoft Windows RPC
139/tcp  open  netbios-ssn      Microsoft Windows netbios-ssn
389/tcp  open  ldap             Microsoft Windows Active Directory LDAP (Domain: thm.local, Site: Default-First-Site-Name)
443/tcp  open  ssl/https?
| tls-alpn:
|   h2
|_  http/1.1
|_ssl-date: 2026-08-26T13:43:27+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=thm-LABYRINTH-CA
| Not valid before: 2023-05-12T07:26:00
|_Not valid after:  2028-05-12T07:35:59
445/tcp  open  microsoft-ds?
464/tcp  open  kpasswd5?
593/tcp  open  ncacn_http       Microsoft Windows RPC over HTTP 1.0
636/tcp  open  ldapssl?
3268/tcp open  ldap             Microsoft Windows Active Directory LDAP (Domain: thm.local, Site: Default-First-Site-Name)
3269/tcp open  globalcatLDAPssl?
3389/tcp open  ms-wbt-server    Microsoft Terminal Services
|_ssl-date: 2026-08-26T13:43:27+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=ad.thm.local
| Not valid before: 2026-08-25T13:18:28
|_Not valid after:  2027-02-24T13:18:28
| rdp-ntlm-info:
|   Target_Name: THM
|   NetBIOS_Domain_Name: THM
|   NetBIOS_Computer_Name: AD
|   DNS_Domain_Name: thm.local
|   DNS_Computer_Name: ad.thm.local
|   Product_Version: 10.0.17763
|_  System_Time: 2026-08-26T13:42:14+00:00
Service Info: Host: AD; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-time:
|   date: 2026-08-26T13:42:16
|_  start_date: N/A
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
```

The Nmap scan shows that multiple ports are opened such that services like Kerberos (88), LDAP (389/636/3268/3269), SMB (445), and DNS (53) are opened.

The presence of Kerberos, LDAP, SMB, and DNS services strongly indicates that the target is a Domain Controller (DC) and identified the hostname as AD and the domain as thm.local.



```
~ via 🐘 v8.4.16 via 🐍 v3.13.12
❯ sudo nano /etc/hosts
[sudo] password for manuel:


~ via 🐘 v8.4.16 via 🐍 v3.13.12 took 19s
❯ nxc smb ad.thm.local -u 'guest' -p '' --shares
SMB         10.81.180.58    445    AD              [*] Windows 10 / Server 2019 Build 17763 x64 (name:AD) (domain:thm.local) (signing:True) (SMBv1:None) (Null Auth:True)
SMB         10.81.180.58    445    AD              [+] thm.local\guest:
SMB         10.81.180.58    445    AD              [*] Enumerated shares
SMB         10.81.180.58    445    AD              Share           Permissions     Remark
SMB         10.81.180.58    445    AD              -----           -----------     ------
SMB         10.81.180.58    445    AD              ADMIN$                          Remote Admin
SMB         10.81.180.58    445    AD              C$                              Default share
SMB         10.81.180.58    445    AD              IPC$            READ            Remote IPC
SMB         10.81.180.58    445    AD              NETLOGON                        Logon server share
SMB         10.81.180.58    445    AD              SYSVOL                          Logon server share


~ via 🐘 v8.4.16 via 🐍 v3.13.12 took 13s
❯ nxc ldap ad.thm.local -u 'guest' -p ''
LDAP        10.81.180.58    389    AD              [*] Windows 10 / Server 2019 Build 17763 (name:AD) (domain:thm.local) (signing:None) (channel binding:No TLS cert)
LDAP        10.81.180.58    389    AD              [+] thm.local\guest:
```

Figure 1.1 shows that the guest account allows authentication without requiring a password and showing the DC is running Windows Server 2019 (Build 17763) with domain thm.local

Impact:

This could allow an attacker to enumerate domain information and identify further attack opportunities

Remediation advises:

Disable the Guest account or require proper authentication and restrict its access to domain resources.

Title: Kerberoastable Account with Weak Password

Risk rating: High (CVSS score-7,8)

Summary:

The CODY_ROY account had an SPN configured, allowing a Kerberos service ticket to be requested, and the account password hash to be obtained for offline cracking.

Background:

An attacker with valid domain access could request a Kerberos service ticket for a Kerberoastable account and attempt to crack the ticket offline. If the password is weak, recovering it could provide valid credentials, which may then be used to access domain resources, enumerate other users, and pursue further privilege escalation.

Technical Details and Evidence:

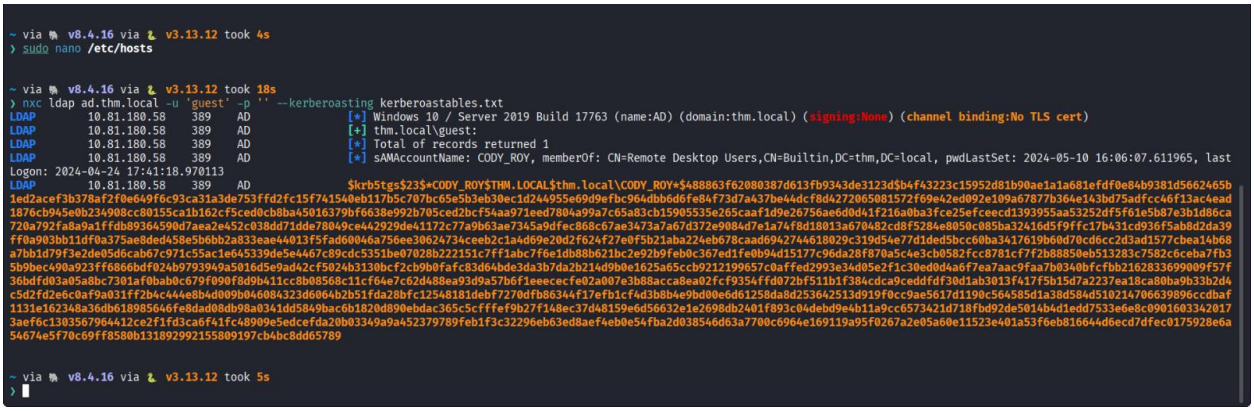


Figure 1.2: Shows the Kerberoasting enumeration, identifying the CODY_ROY account, and extracting its TGS hash.

LDAP enumeration identified CODY_ROY as a Kerberoastable account because it had a Service Principal Name (SPN) configured. A Kerberos TGS ticket was requested, producing a hash that could be cracked offline to recover the account password.

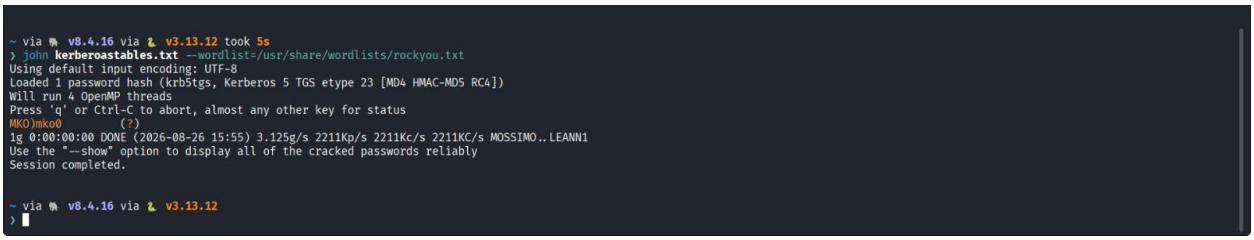


Figure 1.3: Shows the weak password successfully cracked using John the Ripper.

Impact:

The weak password was successfully cracked, providing valid domain credentials that could be used for further enumeration and privilege escalation.

Remediation advises:

Use strong, unique passwords for service accounts and regularly review accounts with SPNs. Where possible, use managed service accounts and stronger Kerberos encryption to reduce the risk of successful offline password cracking.

Title: Credential Management Weakness

Risk rating: critical

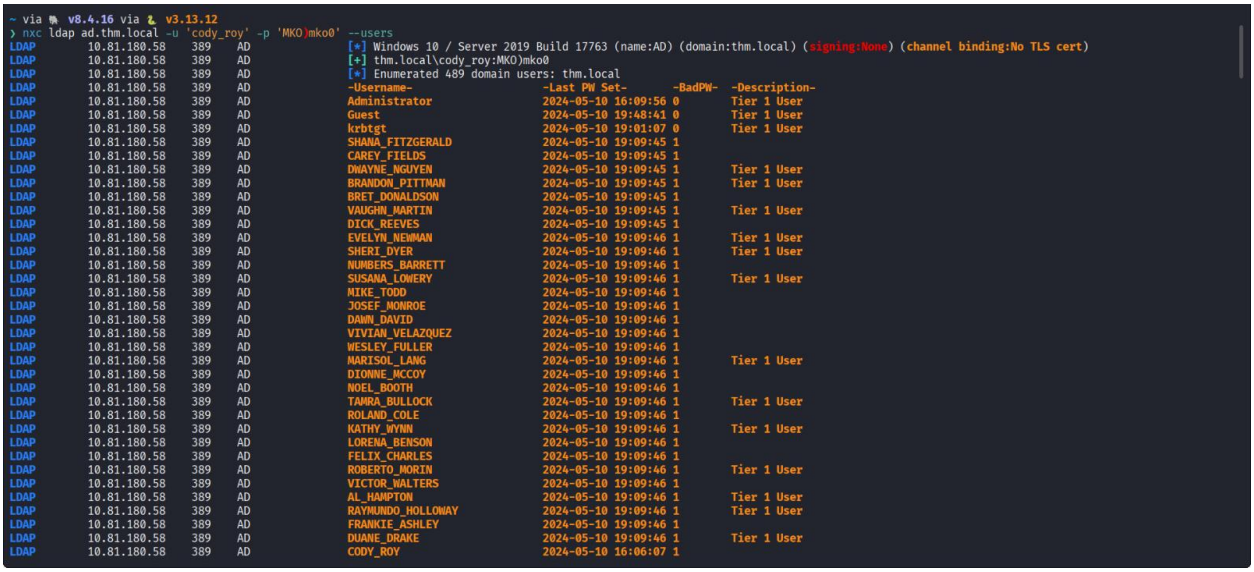
Summary:

The password recovered from CODY_ROY was reused by another domain account, ZACHARY_HUNT. This allowed the compromised password to be used to authenticate as another user.

Background:

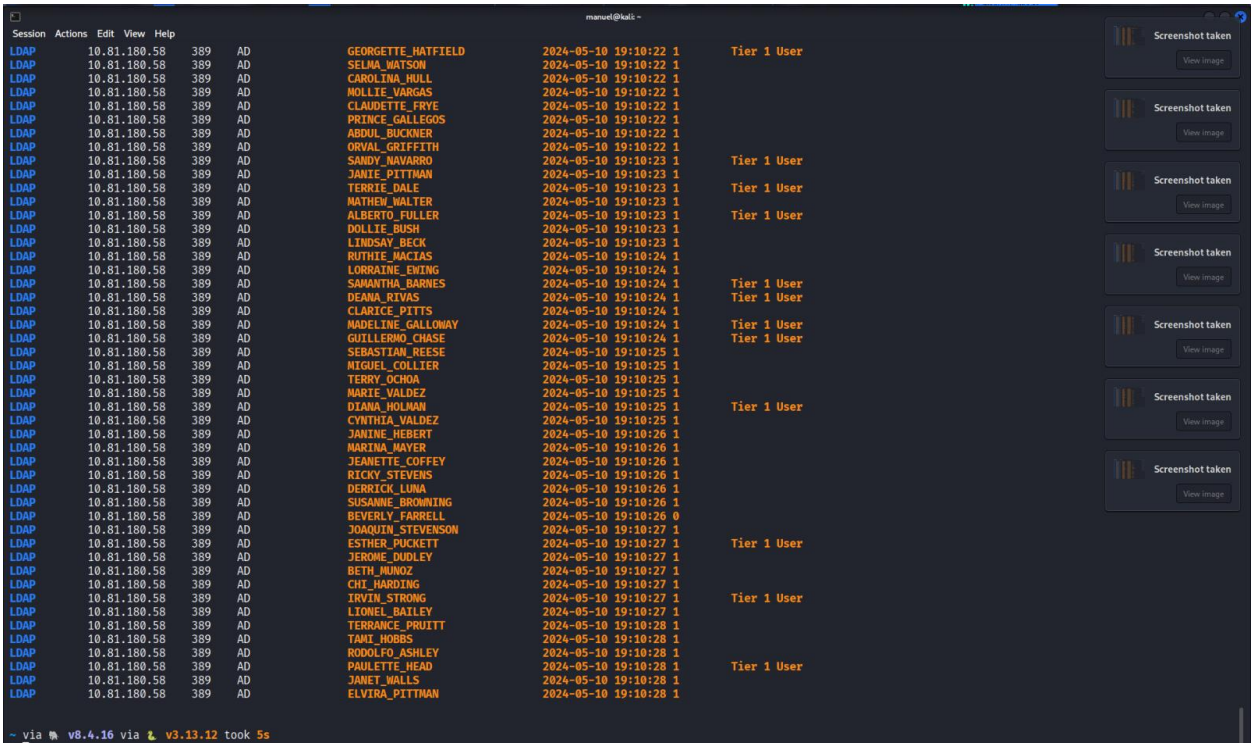
An attacker who obtains a user password may test it against other known usernames. If the password has been reused, they could gain access to additional accounts, expand their access within the domain, and potentially discover higher-privileged accounts or further paths to privilege escalation.

Technical Details and Evidence:



```
~ via 🐘 v8.4.16 via 🐍 v3.13.12
❯ nxc ldap ad.thm.local -u 'cody_roy' -p 'MKO)mko0' --users
LDAP        10.81.180.58    389    AD              [*] Windows 10 / Server 2019 Build 17763 (name:AD) (domain:thm.local) (signing:None) (channel binding:No TLS cert)
LDAP        10.81.180.58    389    AD              [+] thm.local\cody_roy:MKO)mko0
LDAP        10.81.180.58    389    AD              [*] Enumerated 489 domain users: thm.local
LDAP        10.81.180.58    389    AD              -Username-                -Last PW Set-       -BadPW-  -Description-
LDAP        10.81.180.58    389    AD              Administrator             2024-05-10 16:09:56 0        Tier 1 User
LDAP        10.81.180.58    389    AD              Guest                     2024-05-10 19:48:41 0        Tier 1 User
LDAP        10.81.180.58    389    AD              krbtgt                    2024-05-10 19:01:07 0        Tier 1 User
LDAP        10.81.180.58    389    AD              SHANA_FITZGERALD          2024-05-10 19:09:45 1
LDAP        10.81.180.58    389    AD              CAREY_FIELDS              2024-05-10 19:09:45 1
LDAP        10.81.180.58    389    AD              DWAYNE_NGUYEN             2024-05-10 19:09:45 1        Tier 1 User
LDAP        10.81.180.58    389    AD              BRANDON_PITTMAN           2024-05-10 19:09:45 1        Tier 1 User
LDAP        10.81.180.58    389    AD              BRET_DONALDSON            2024-05-10 19:09:45 1
LDAP        10.81.180.58    389    AD              VAUGHN_MARTIN             2024-05-10 19:09:45 1        Tier 1 User
LDAP        10.81.180.58    389    AD              DICK_REEVES               2024-05-10 19:09:45 1
LDAP        10.81.180.58    389    AD              EVELYN_NEWMAN             2024-05-10 19:09:46 1        Tier 1 User
LDAP        10.81.180.58    389    AD              SHERI_DYER                2024-05-10 19:09:46 1        Tier 1 User
LDAP        10.81.180.58    389    AD              NUMBERS_BARRETT           2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              SUSANA_LOWERY             2024-05-10 19:09:46 1        Tier 1 User
LDAP        10.81.180.58    389    AD              MIKE_TODD                 2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              JOSEF_MONROE              2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              DAWN_DAVID                2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              VIVIAN_VELAZQUEZ          2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              WESLEY_FULLER             2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              MARISOL_LANG              2024-05-10 19:09:46 1        Tier 1 User
LDAP        10.81.180.58    389    AD              DIONNE_MCCOY              2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              NOEL_BOOTH                2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              TAMRA_BULLOCK             2024-05-10 19:09:46 1        Tier 1 User
LDAP        10.81.180.58    389    AD              ROLAND_COLE               2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              KATHY_WYNN                2024-05-10 19:09:46 1        Tier 1 User
LDAP        10.81.180.58    389    AD              LORENA_BENSON             2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              FELIX_CHARLES             2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              ROBERTO_MORIN             2024-05-10 19:09:46 1        Tier 1 User
LDAP        10.81.180.58    389    AD              VICTOR_WALTERS            2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              AL_HAMPTON                2024-05-10 19:09:46 1        Tier 1 User
LDAP        10.81.180.58    389    AD              RAYMUNDO_HOLLOWAY         2024-05-10 19:09:46 1        Tier 1 User
LDAP        10.81.180.58    389    AD              FRANCIE_ASHLEY            2024-05-10 19:09:46 1
LDAP        10.81.180.58    389    AD              DUANE_DRAKE               2024-05-10 19:09:46 1        Tier 1 User
LDAP        10.81.180.58    389    AD              CODY_ROY                  2024-05-10 16:06:07 1
```

Figure 1.4 shows enumeration domain users with NetExec ().



```
Session  Actions  Edit  View  Help                                  manuel@kali: ~
LDAP        10.81.180.58    389    AD              GEORGETTE_HATFIELD        2024-05-10 19:10:22 1        Tier 1 User
LDAP        10.81.180.58    389    AD              SELMA_WATSON              2024-05-10 19:10:22 1
LDAP        10.81.180.58    389    AD              CAROLINA_HULL             2024-05-10 19:10:22 1
LDAP        10.81.180.58    389    AD              MOLLIE_VARGAS             2024-05-10 19:10:22 1
LDAP        10.81.180.58    389    AD              CLAUDETTE_FRYE            2024-05-10 19:10:22 1
LDAP        10.81.180.58    389    AD              PRINCE_GALLEGOS           2024-05-10 19:10:22 1
LDAP        10.81.180.58    389    AD              ABDUL_BUCKNER             2024-05-10 19:10:22 1
LDAP        10.81.180.58    389    AD              ORVAL_GRIFFITH            2024-05-10 19:10:22 1
LDAP        10.81.180.58    389    AD              SANDY_NAVARRO             2024-05-10 19:10:23 1        Tier 1 User
LDAP        10.81.180.58    389    AD              JANIE_PITTMAN             2024-05-10 19:10:23 1
LDAP        10.81.180.58    389    AD              TERRIE_DALE               2024-05-10 19:10:23 1        Tier 1 User
LDAP        10.81.180.58    389    AD              MATHEW_WALTER             2024-05-10 19:10:23 1
LDAP        10.81.180.58    389    AD              ALBERTO_FULLER            2024-05-10 19:10:23 1        Tier 1 User
LDAP        10.81.180.58    389    AD              DOLLIE_BUSH               2024-05-10 19:10:23 1
LDAP        10.81.180.58    389    AD              LINDSAY_BECK              2024-05-10 19:10:23 1
LDAP        10.81.180.58    389    AD              RUTHIE_MACIAS             2024-05-10 19:10:24 1
LDAP        10.81.180.58    389    AD              LORRAINE_EWING            2024-05-10 19:10:24 1
LDAP        10.81.180.58    389    AD              SAMANTHA_BARNES           2024-05-10 19:10:24 1        Tier 1 User
LDAP        10.81.180.58    389    AD              DEANA_RIVAS               2024-05-10 19:10:24 1        Tier 1 User
LDAP        10.81.180.58    389    AD              CLARICE_PITTS             2024-05-10 19:10:24 1
LDAP        10.81.180.58    389    AD              MADELINE_GALLOWAY         2024-05-10 19:10:24 1        Tier 1 User
LDAP        10.81.180.58    389    AD              GUILLERMO_CHASE           2024-05-10 19:10:24 1        Tier 1 User
LDAP        10.81.180.58    389    AD              SEBASTIAN_REESE           2024-05-10 19:10:25 1
LDAP        10.81.180.58    389    AD              MIGUEL_COLLIER            2024-05-10 19:10:25 1
LDAP        10.81.180.58    389    AD              TERRY_OCHOA               2024-05-10 19:10:25 1
LDAP        10.81.180.58    389    AD              MARIE_VALDEZ              2024-05-10 19:10:25 1
LDAP        10.81.180.58    389    AD              DIANA_HOLMAN              2024-05-10 19:10:25 1        Tier 1 User
LDAP        10.81.180.58    389    AD              CYNTHIA_VALDEZ            2024-05-10 19:10:25 1
LDAP        10.81.180.58    389    AD              JANINE_HEBERT             2024-05-10 19:10:26 1
LDAP        10.81.180.58    389    AD              MARINA_MAYER              2024-05-10 19:10:26 1
LDAP        10.81.180.58    389    AD              JEANETTE_COFFEY           2024-05-10 19:10:26 1
LDAP        10.81.180.58    389    AD              RICKY_STEVENS             2024-05-10 19:10:26 1
LDAP        10.81.180.58    389    AD              DERRICK_LUNA              2024-05-10 19:10:26 1
LDAP        10.81.180.58    389    AD              SUSANNE_BROWNING          2024-05-10 19:10:26 1
LDAP        10.81.180.58    389    AD              BEVERLY_FARRELL           2024-05-10 19:10:26 0
LDAP        10.81.180.58    389    AD              JOAQUIN_STEVENSON         2024-05-10 19:10:27 1
LDAP        10.81.180.58    389    AD              ESTHER_PUCKETT            2024-05-10 19:10:27 1        Tier 1 User
LDAP        10.81.180.58    389    AD              JEROME_DUDLEY             2024-05-10 19:10:27 1
LDAP        10.81.180.58    389    AD              BETH_MUNOZ                2024-05-10 19:10:27 1
LDAP        10.81.180.58    389    AD              CHI_HARDING               2024-05-10 19:10:27 1
LDAP        10.81.180.58    389    AD              IRVIN_STRONG              2024-05-10 19:10:27 1        Tier 1 User
LDAP        10.81.180.58    389    AD              LIONEL_BAILEY             2024-05-10 19:10:27 1
LDAP        10.81.180.58    389    AD              TERRANCE_PRUITT           2024-05-10 19:10:28 1
LDAP        10.81.180.58    389    AD              TAMI_HOBBS                2024-05-10 19:10:28 1
LDAP        10.81.180.58    389    AD              RODOLFO_ASHLEY            2024-05-10 19:10:28 1
LDAP        10.81.180.58    389    AD              PAULETTE_HEAD             2024-05-10 19:10:28 1        Tier 1 User
LDAP        10.81.180.58    389    AD              JANET_WALLS               2024-05-10 19:10:28 1
LDAP        10.81.180.58    389    AD              ELVIRA_PITTMAN            2024-05-10 19:10:28 1

~ via 🐘 v8.4.16 via 🐍 v3.13.12 took 5s
```

Figure 1.5: Show a list of users.

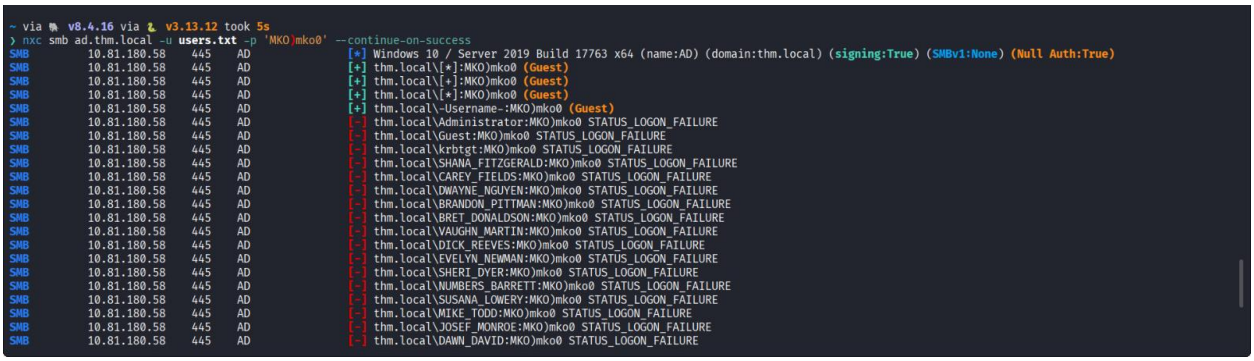
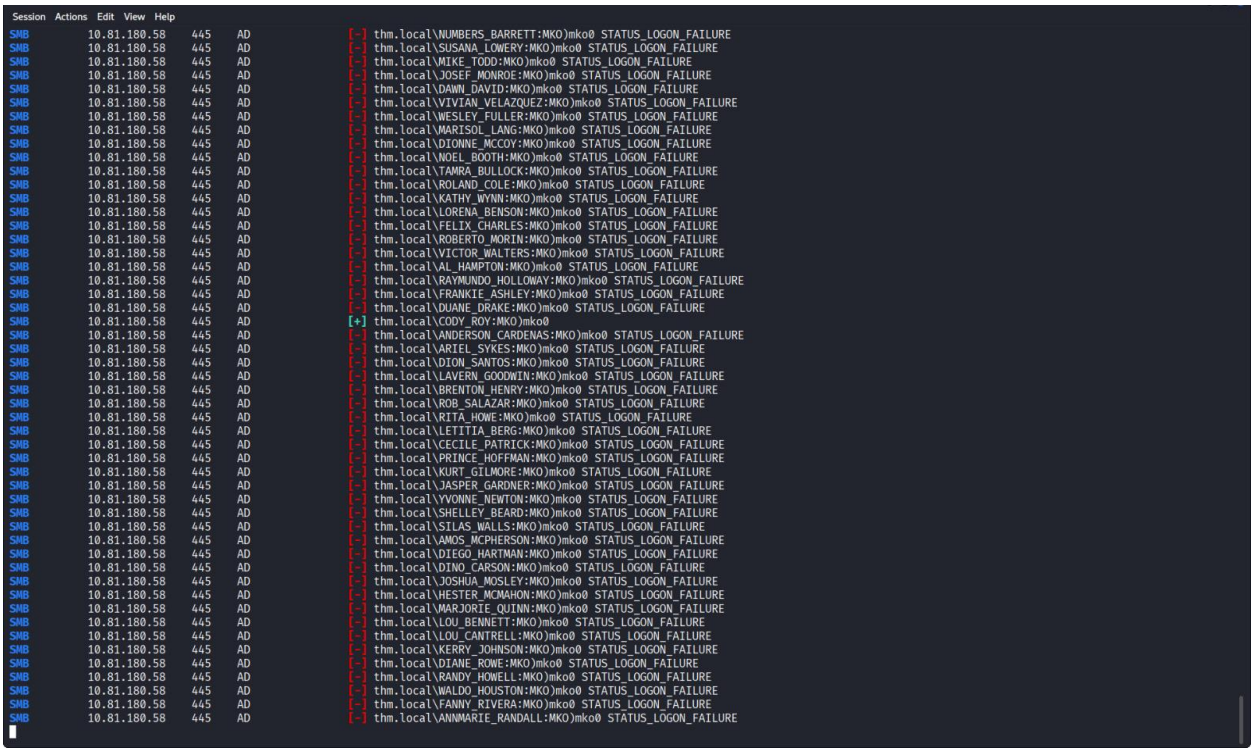


Figure 1.6: shows password spraying on every user.



Shows successful authentication to ZACHARY_HUNT using the password recovered from CODY_ROY.

Impact:

Password reuse increased the impact of the initial compromise and provided the attacker with access to another domain account, enabling further enumeration and privilege escalation.

Remediation advises:

Enforce unique passwords for all domain accounts and prevent users from reusing passwords across accounts. Use strong password policies and monitor compromised credentials to limit the impact of a single account being compromised.

Title: Active Directory Access Control Misconfiguration

Risk rating: High

Summary:

ZACHARY_HUNT was granted GenericWrite permissions over JERRI_LANCASTER. This excessive permission allowed the account to modify attributes of the target account and enabled a targeted Kerberoasting attack.

Background:

An attacker who compromises ZACHARY_HUNT could abuse its excessive GenericWrite permissions over JERRI_LANCASTER to modify the account and perform targeted Kerberoasting. If the resulting password is weak, the attacker could recover JERRI_LANCASTER's credentials and use the account to gain further access within the domain.

Technical Details and Evidence:

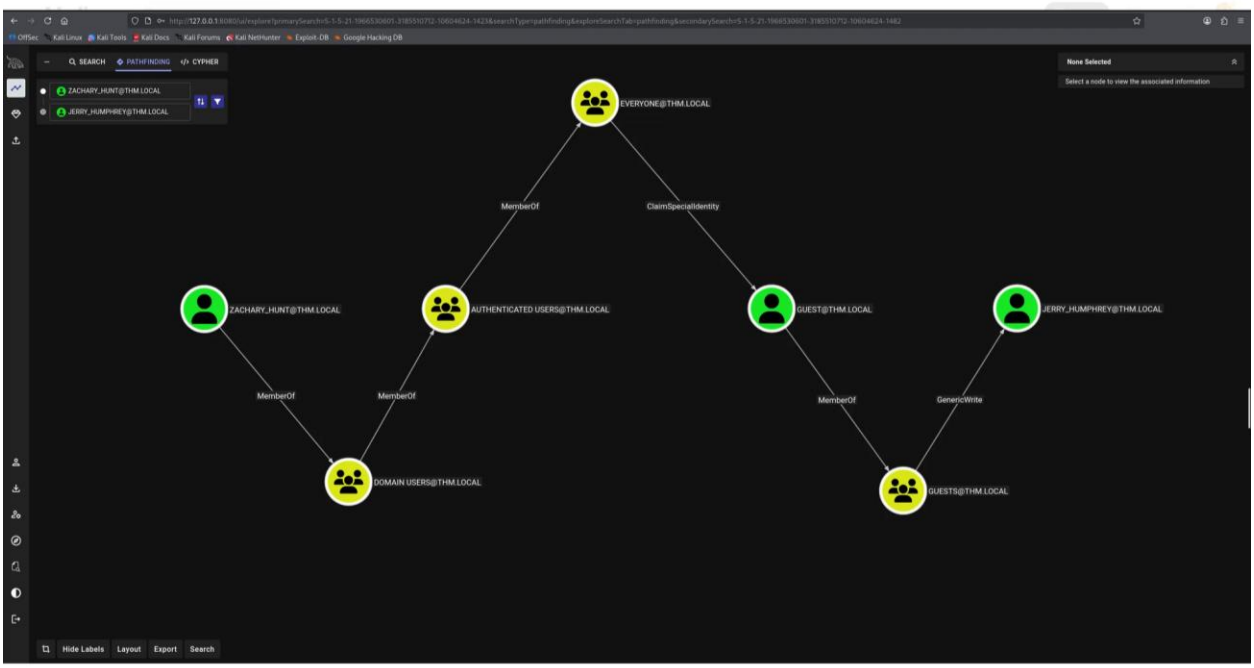


Figure 1.7: Shows the GenericWrite relationship between ZACHARY_HUNT and JERRI_LANCASTER.

Impact:

An attacker controlling ZACHARY_HUNT could abuse the permission to obtain a Kerberos service ticket for JERRI_LANCASTER and attempt offline password cracking, potentially leading to further account compromise.

Remediation advises:

Remove the unnecessary GenericWrite permission from ZACHARY_HUNT and apply the principle of least privilege. Regularly review Active Directory permissions to ensure users only have the access required for their roles.

Title: Plaintext Credentials in PowerShell Script

Risk rating: high

Summary:

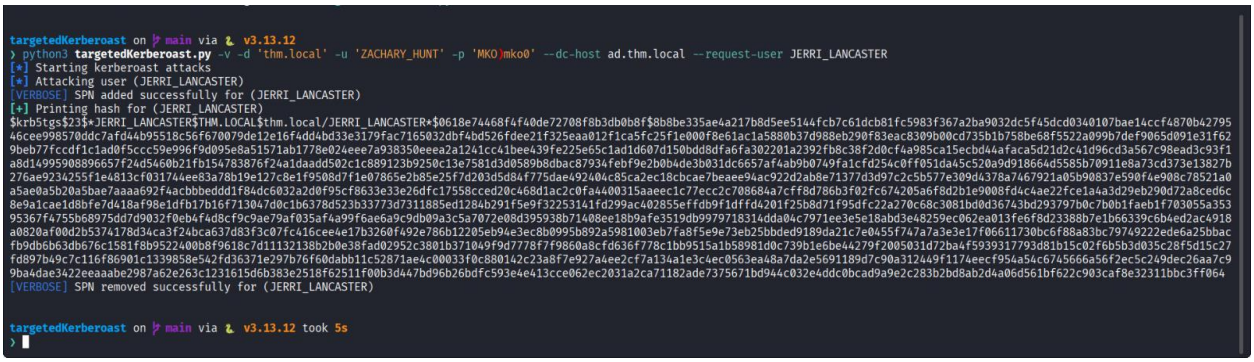
A PowerShell script (syncer.ps1) on the system contains plaintext credentials for the SANFORD_DAUGHERTY account.

Background:

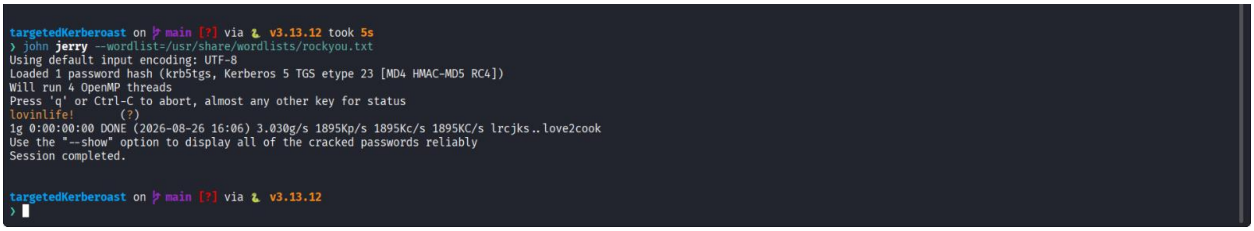
An attacker who gains access to the script could retrieve the exposed credentials and authenticate as SANFORD_DAUGHERTY, potentially gaining additional privileges depending on the account's permissions.

Technical Details and Evidence:

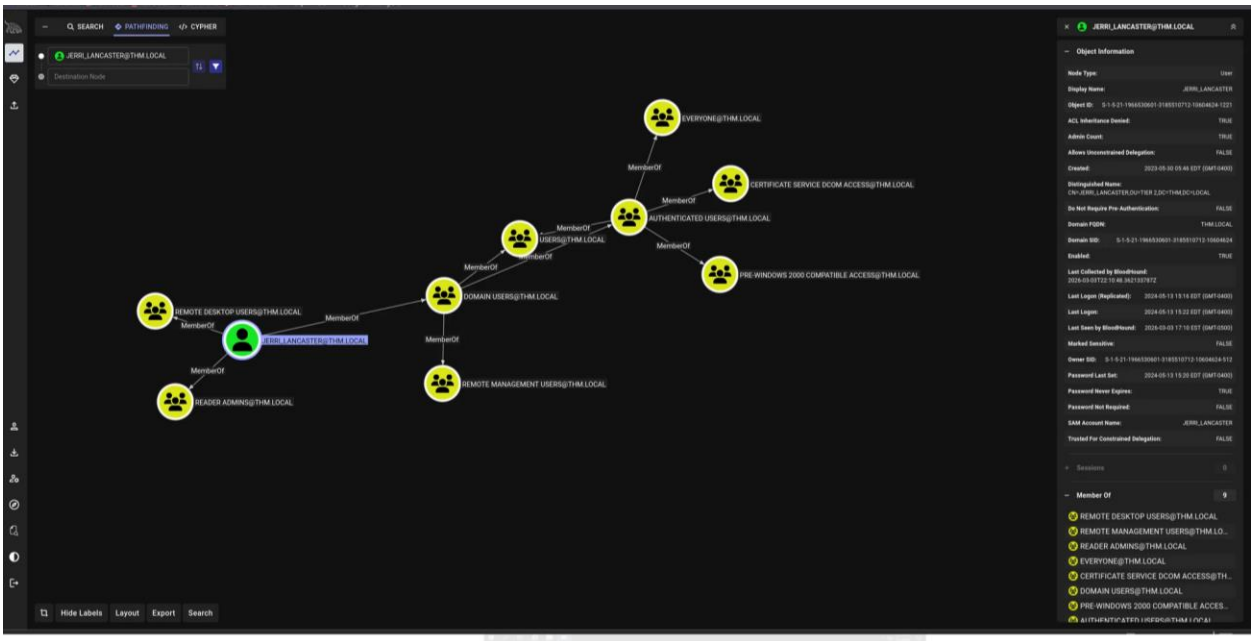
Using the credentials obtained for ZACHARY_HUNT, a temporary Service Principal Name (SPN) was assigned to JERRI_LANCASTER. A Kerberos service ticket was then requested for the account, resulting in a Kerberoastable TGS hash.



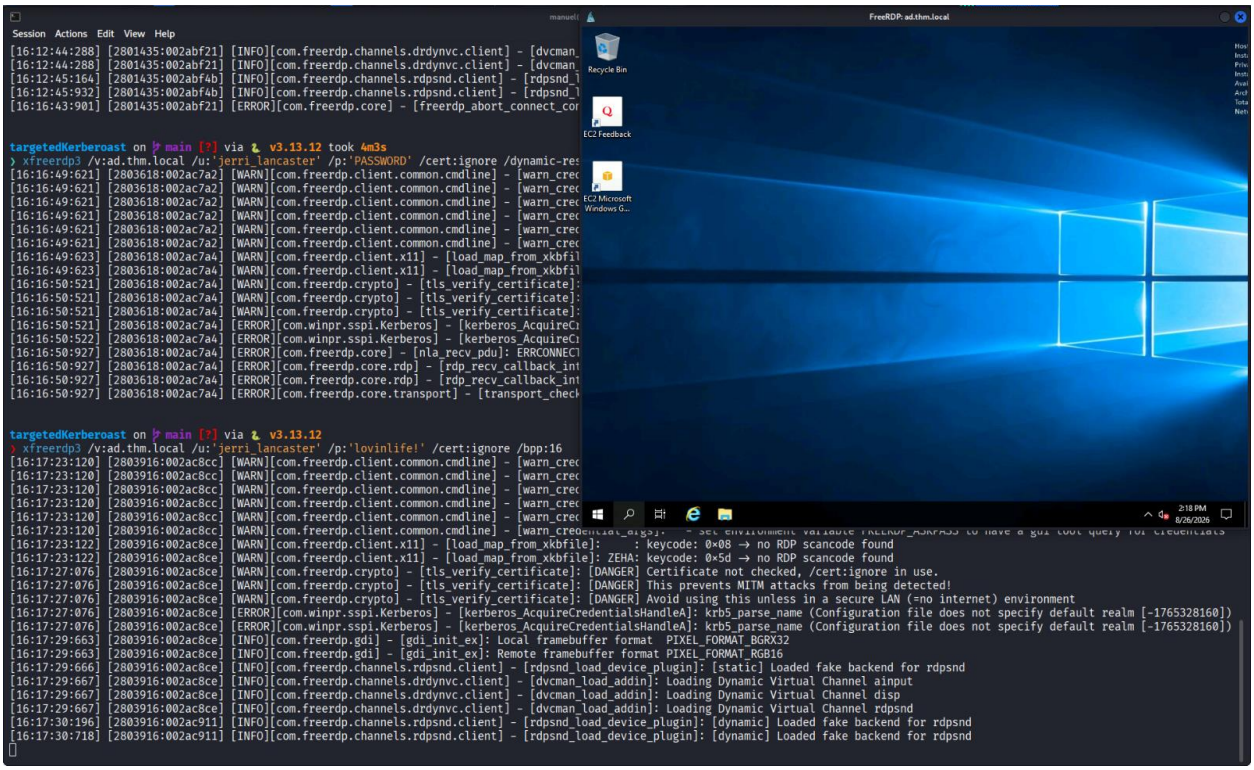
The extracted hash was saved and processed using John the Ripper with the rockyou.txt wordlist. The password was successfully cracked as lovinlife!.



Further enumeration with BloodHound showed that JERRI_LANCASTER was a member of the Remote Desktop Users group. .

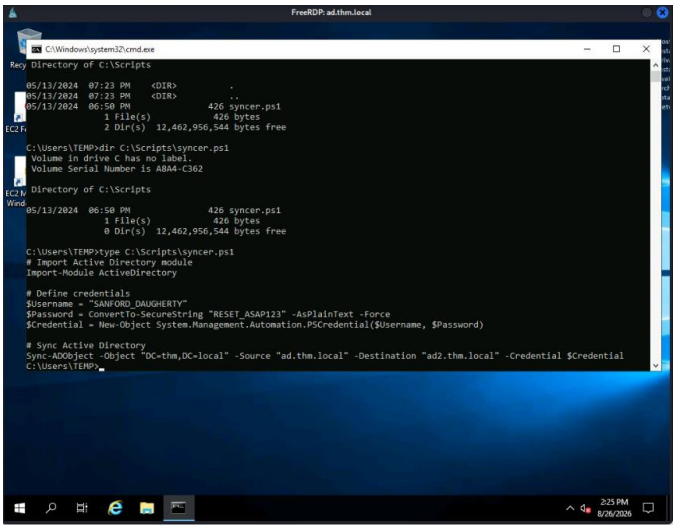


This membership provided the ability to authenticate to the target system through Remote Desktop Protocol (RDP) using the recovered credentials.



After gaining RDP access as JERRI_LANCASTER, the Windows filesystem was enumerated using the command prompt. An interesting directory named C:\Scripts was identified.

Further investigation of this directory revealed a PowerShell script named syncer.ps1



Reviewing the contents of the script exposed credentials belonging to the SANFORD_DAUGHERTY account in plaintext.

This finding demonstrates how storing credentials directly within scripts can allow an attacker access to the system to recover valid credentials and potentially escalate their privileges.

Impact:

The exposed SANFORD_DAUGHERTY credentials provided access to a local administrator accoun**t**, allowing the attacker to escalate privileges and ultimately obtain SYSTEM-level access.

Remediation advises:

Remove all plaintext credentials from syncer.ps1 and store them securely using a proper credential-management solution. The exposed SANFORD_DAUGHERTY password should also be changed immediately, and access to sensitive scripts should be restricted to authorized users only.

Title: Excessive Local Administrator Privileges

Risk rating: Critical

Summary:

The SANFORD_DAUGHERTY account was configured with local administrator privileges on the target system. After discovering the account credentials in syncer.ps1, authentication was successfully verified, confirming administrative access.

Background:

An attacker who obtains credentials for a local administrator account can perform privileged actions on the system, including accessing protected files, modifying system configurations, and executing processes with elevated privileges.

Technical Details and Evidence:

The credentials discovered in syncer.ps1 were tested against the Domain Controller using NetExec. The authentication was successful, and the (Pwn3d!) result indicated that SANFORD_DAUGHERTY had administrative privileges on the target system.

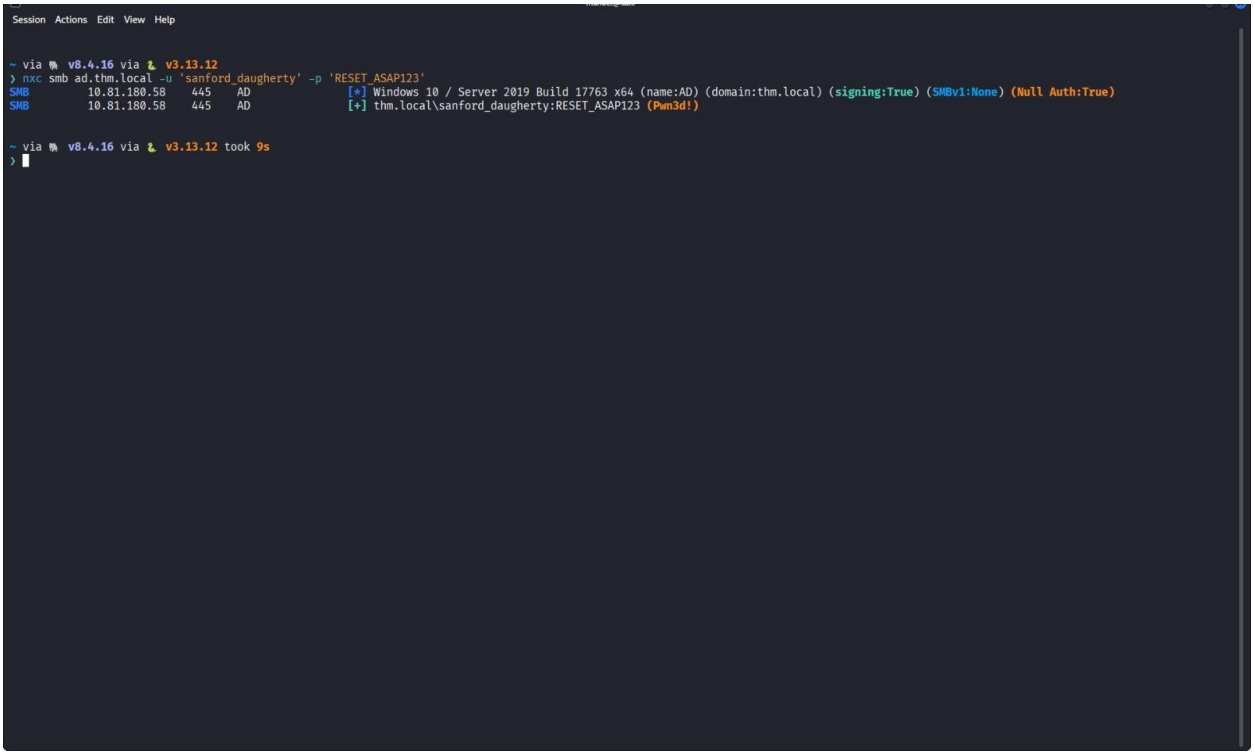


Figure 1.8: Shows successful authentication as SANFORD_DAUGHERTY with administrative privileges.

The administrative access was then leveraged through Impacket's smbexec functionality to obtain a system level shell. From this privileged context, the attacker was able to access protected files, including the flag stored at:

C:\Users\Administrator\Desktop\flag.txt.txt



Figure 1.9: Shows the use of SMB-based remote execution to obtain SYSTEM-level access and retrieve the flag.

Impact:

The administrator privileges allowed the attacker to use SMB-based remote execution to obtain a system level shell, resulting in complete control of the target machine and access to the final flag.

Remediation advises:

Remove unnecessary local administrator privileges from SANFORD_DAUGHERTY and apply least-privilege access. The exposed credentials should be rotated immediately, and administrative accounts should be restricted and closely monitored. Remote administrative services should also be limited to authorized users and systems.